

Python Tutorial Notebook

Strings, String Functions

Topics covered so far:

- 1) `print()` and `input()`
- 2) Variables and Data Types
- 3) Conditionals and Strings

This notebook reviews **strings** with examples, then gives you **practice tasks** (no advanced concepts like loops/functions/lists required).

Strings in Python

A **string** is text inside quotes: "Hello" or 'Hello'.

In [1]:

```
name = "Arif"
city = 'Sukkur'
sentence = "Python is easy!"
print(name, city, sentence)
```

Out [1]:

```
Arif Sukkur Python is easy!
```

Multi-line strings

Use triple quotes for multiple lines.

In [2]:

```
text = """This is line 1
This is line 2
This is line 3
Hello There"""
print(text)
```

Out [2]:

```
This is line 1
This is line 2
This is line 3
Hello There
```

Indexing (access single characters)

Index starts at 0. Negative indexing starts from the end.

In [3]:

```
word = "PYTHON"
print(word[0])    # P
print(word[3])    # H
print(word[-1])   # N
```

Out [3]:

P
H
N

Slicing (get a part of a string)

Format: s[start:end:step]

In [4]:

```
word = "PYTHON"
print(word[0:3])    # PYT
print(word[2:])     # THON
print(word[:4])     # PYTH
print(word[::2])    # PTO
print(word[::-1])   # NOHTYP
```

Out [4]:

PYT
THON
PYTH
PTO
NOHTYP

Strings are immutable

You can't change a character directly. Create a new string instead.

In [5]:

```
s = "cat"
s = "b" + s[1:]
print(s)  # bat
```

Out [5]:

bat

Part C — Common String Methods (Functions)

Most string functions are **methods**: text.method()

1) Case conversion

- upper() → uppercase
- lower() → lowercase
- title() → Title Case
- capitalize() → first letter capital

In [6]:

```
s = "python programming"
print(s.upper())
print(s.lower())
print(s.title())
print(s.capitalize())
```

Out [6]:

```
PYTHON PROGRAMMING
python programming
Python Programming
Python programming
```

2) Removing spaces

- `strip()` removes spaces from both sides
- `rstrip()` from left
- `rstrip()` from right

In [7]:

```
s = "  hello  "
print(s.strip())
print(s.lstrip())
print(s.rstrip())
```

Out [7]:

```
hello
hello
  hello
```

3) Searching and counting

- `find()` returns position or -1
- `count()` counts occurrences

In [8]:

```
s = "I love Python"
print(s.find("Python")) # 7
print(s.find("Java"))   # -1
print("banana".count("a"))
```

Out [8]:

```
7
-1
3
```

4) Replace text

- `replace(old, new)`

In [9]:

```
s = "I like tea"
print(s.replace("tea", "coffee"))
```

Out [9]:

```
I like coffee
```

5) Checking start/end

- `startswith()`
- `endswith()`

In [10]:

```
s = "report.pdf"
print(s.startswith("rep"))
print(s.endswith(".pdf"))
```

Out [10]:

```
True
True
```

6) Checking the type of text

- `isalpha()` letters only
- `isdigit()` digits only
- `isalnum()` letters+digits
- `isspace()` spaces only

In [11]:

```
print("Hello".isalpha())
print("123".isdigit())
print("A1".isalnum())
print("   ".isspace())
```

Out [11]:

```
True
True
True
True
```

7) String formatting (recommended)

Use f-strings:

In [12]:

```
name = "Arif"
age = 24
print(f"My name is {name} and I am {age} years old.")
```

Out [12]:

```
My name is Arif and I am 24 years old.
```

Practice Tasks (Strings + Conditionals)

Rules:

- Use only: `input()`, `print()`, variables, string methods, indexing/slicing, `if/elif/else`
- Do NOT use loops, lists, functions, files, or advanced modules.

Try each task by writing your code in the empty cell under it.

Task 1 — Uppercase Name

Take a name from the user and print it in UPPERCASE.

Example: input: arif → output: ARIF

In [13]:

```
# Write your solution here
```

Task 2 — Greeting with Length

Take a name and print: Hello ! Your name has characters.

```
In [14]:
```

```
# Write your solution here
```

Task 3 — First and Last Character

Take a word and print its first and last character.

Hint: use indexing word[0] and word[-1].

```
In [15]:
```

```
# Write your solution here
```

Task 4 — Reverse a Word

Take a word and print it reversed.

Hint: slicing [: :-1].

```
In [16]:
```

```
# Write your solution here
```

Task 5 — Trim Spaces

Take a sentence (user may enter extra spaces). Print the cleaned sentence using strip().

```
In [17]:
```

```
# Write your solution here
```

Task 6 — Fever Check

Take temperature as an integer. If > 37 print Fever, else print Normal.

```
In [18]:
```

```
# Write your solution here
```

Task 7 — Even or Odd

Take an integer number. If it is even print Even, else print Odd.

```
In [19]:
```

```
# Write your solution here
```

Task 8 — Grade System

Take marks (0–100). Print grade:

A+ (>=90), A (>=80), B (>=70), C (>=60), else Fail.

```
In [20]:
```

```
# Write your solution here
```

Task 9 — Simple Email Check

Take an email string. If it contains @ and ends with .com, print Valid, else Invalid.

```
In [21]:
```

```
# Write your solution here
```

Task 10 — Starts With 'A'?

Take a name. If it starts with 'A' or 'a', print Starts with A, else print Does not start with A.

```
In [22]:
```

```
# Write your solution here
```

Task 11 — Text Type Checker

Take input text. If it is only digits print Digits only; elif only letters print Letters only; else print Mixed.

```
In [23]:
```

```
# Write your solution here
```

Task 12 — Password Length Rule

Take a password. If length >= 8 print Strong (length OK), else print Weak (too short).

```
In [24]:
```

```
# Write your solution here
```

Optional Challenge (still beginner-friendly)

Combine strings + conditionals:

Challenge: Ask user for a city name. If the user enters an empty string (just presses Enter), print City not provided. Otherwise print City: .

```
In [25]:
```

```
# Write your solution here
```